

Automated Audio Analysis and Podcast Intelligence System

Submitted by : HEMNATH SINGH K

Institution : KINGSTON ENGINEERING COLLEGE

Department : INFORMATION TECHNOLOGY



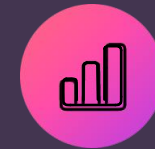
Project Overview



AI-powered system for
analyzing long audio files



Converts podcasts,
interviews, and lectures
into structured insights



Provides transcripts,
topics, summaries, and
emotion analysis



Problem Statement

Long audio content is difficult to search

Users must listen fully to find specific information

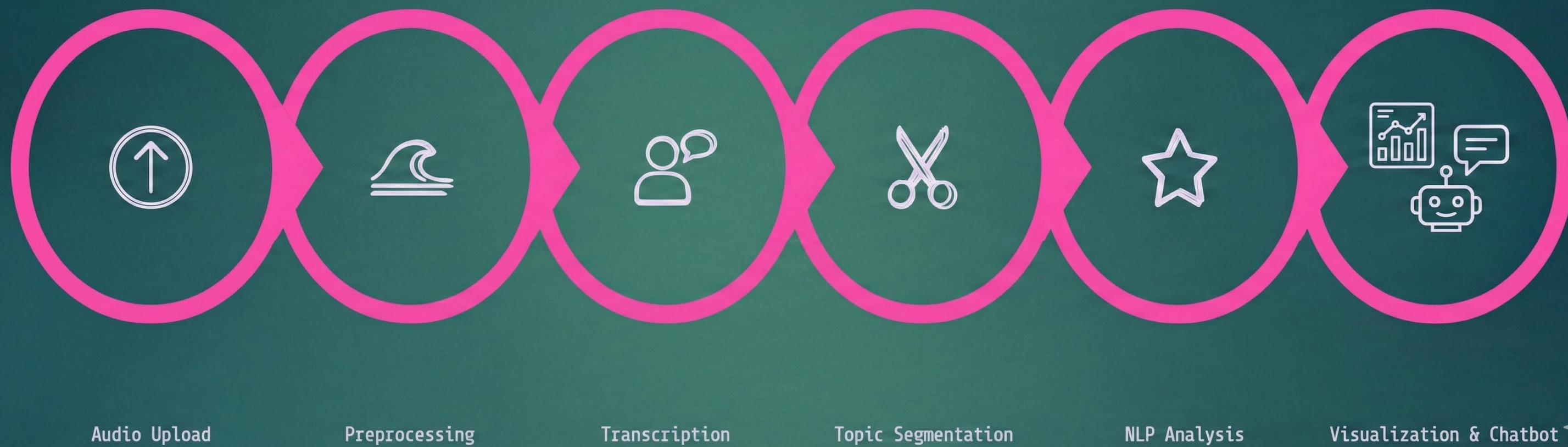
No topic-wise navigation or summaries

Manual analysis is time-consuming and inefficient

Objectives

- 1 Automatic speech-to-text transcription
- 2 Topic-based segmentation of audio
- 3 Keyword extraction and summarization
- 4 Sentiment and emotion analysis
- 5 Interactive user interface for navigation

System Workflow



Methodology

Pipeline-based modular architecture

Each module performs a specific task

Improves scalability and maintainability

Steps:

1. Audio Preprocessing
2. Speech-to-Text Transcription
3. Topic Segmentation
4. Keyword Extraction
5. Summarization
6. Sentiment & Emotion Analysis
7. Indexing and Visualization

Audio Preprocessing

Noise reduction



Volume normalization



Resampling

Purpose:

- Improves transcription accuracy

Tools:

- Librosa
- PyDub
- FFmpeg

Speech-to-Text Transcription

1

Converts speech into text

2

Generates timestamps

3

Supports speaker
diarization

Model:

- Whisper / Faster-Whisper

Output:

- JSON / Text transcripts

Topic Segmentation



Divides transcript into
meaningful topics



Uses semantic similarity
Between sentences



Enables quick navigation of
long audio

Technique:

- Sentence embeddings
- Cosine similarity



Keyword Extraction & Summarization

Extracts important
keywords per topic

Generates summaries and
titles

Reduces long text into
concise insights

Models:

- KeyBERT / YAKE
- DistilBART

Sentiment & Emotion Analysis

Detects sentiment:

Positive, Negative, Neutral

Identifies emotions

such as joy, sadness, anger, fear

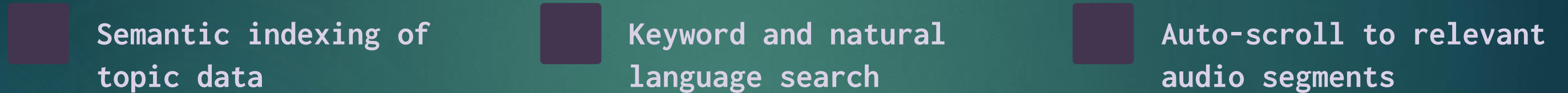
**Works at topic and
segment level**

Use case:

- Understanding speaker mood and tone



Indexing & Search



Benefit:

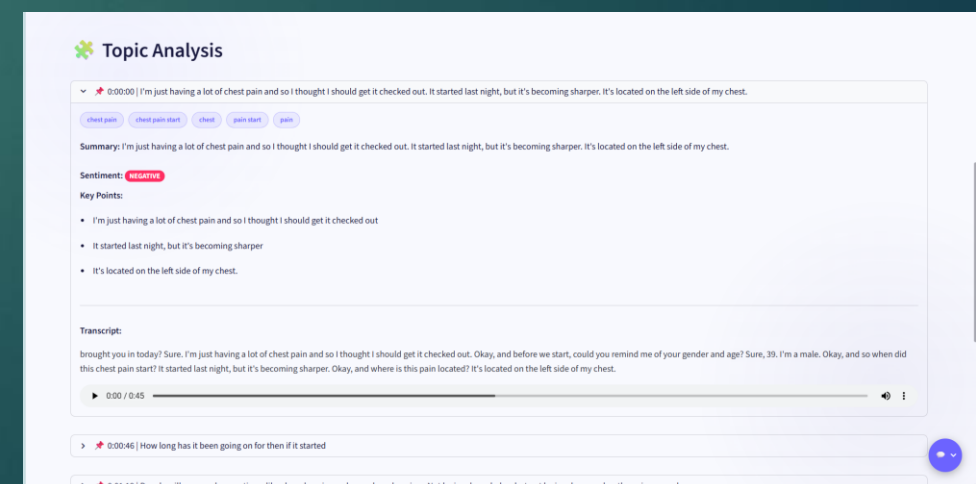
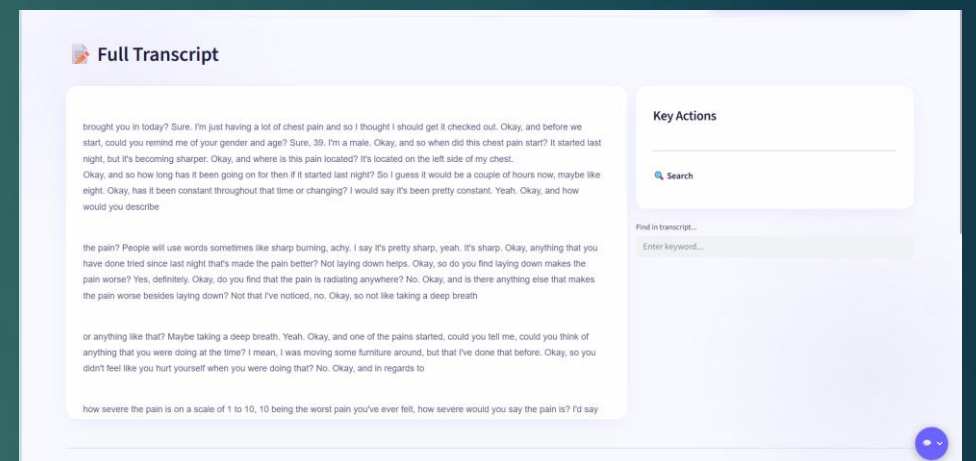
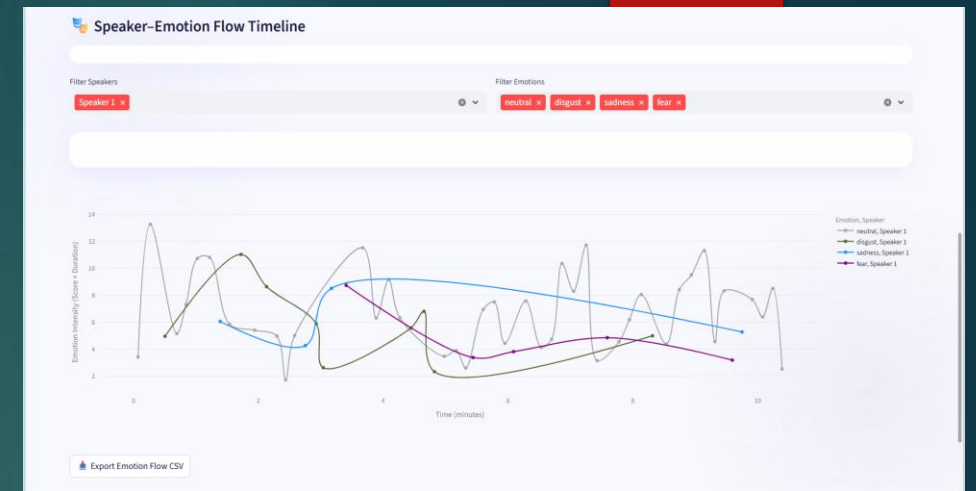
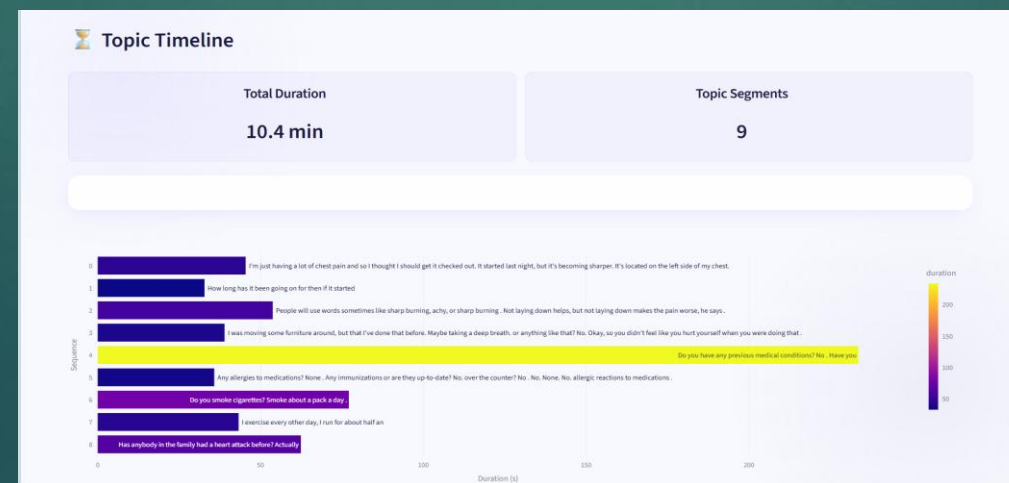
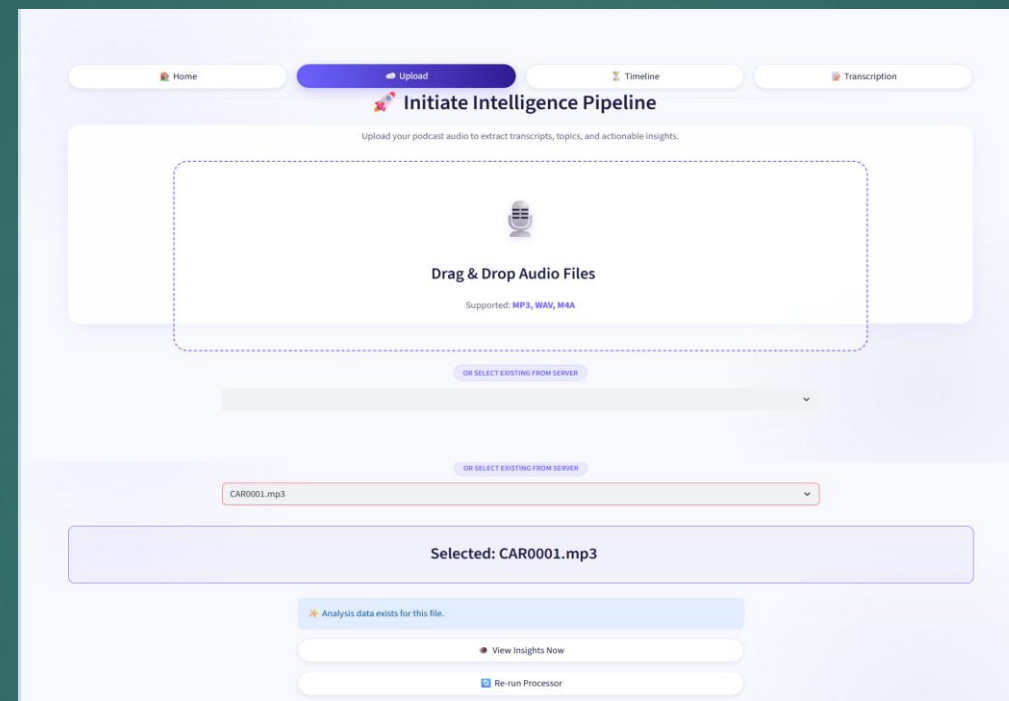
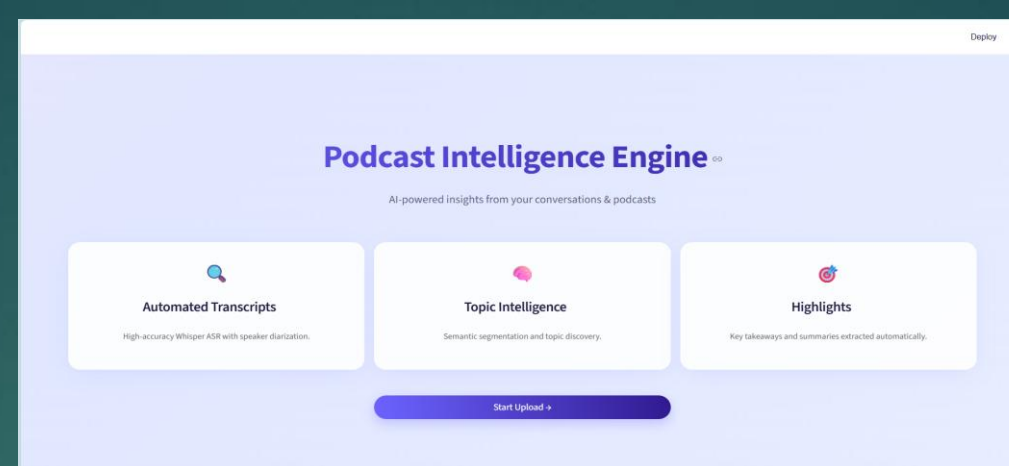
- Faster information retrieval

User Interface

- Built using Streamlit
- Audio upload
- Transcript view
- Topic timeline
- Emotion visualization
- AI chatbot interaction

Demo Video

link





Technology Stack

Audio Processing:

Librosa, PyDub, FFmpeg

Speech-to-Text:

Whisper

NLP:

Sentence Transformers, BERT

Summarization:

DistilBART

UI:

Streamlit

Language:

Python

Challenges

→ Noisy and low-quality audio

→ Accurate topic boundary detection

→ Large audio processing time

→ Emotion detection complexity

Limitations



Performance depends on audio quality



Limited multilingual support



Processing time increases with long audio



Subtle emotions may not be detected accurately

Future Enhancements



Multilingual transcription



Real-time audio analysis



Improved speaker diarization



Cloud deployment



PDF / report export

Use Cases



Educational lectures



Podcast analysis



Medical interviews
(non-diagnostic)



Corporate meetings



Research and analytics

Conclusion

- Successfully built an AI-powered audio analysis system
- Converts unstructured audio into meaningful insights
- Saves time and improves accessibility
- Demonstrates practical use of NLP and speech technologies



Thank You

Questions & Discussion